

Magnetic Equivalent Circuit for Surface-Mounted PM Motor

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Abstract—Electric machine design optimisation requires tools that provide reliable information about machine parameters. Magnetic equivalent circuit (MEC) technique is selected here for predicting machine performance. MEC is developed for a fractional slot, permanent magnet synchronous motor (PMSM). The model includes saturation, fringing effect in air gap and accounts for relative motion of the rotor. Even with saturation being incorporated, the adopted solution converges the model within few iterations. Performance quantities of the motor like flux density, induced EMF, flux linkage, inductance and cogging torque are evaluated. The results of MEC are validated with finite element analysis (FEA) and few with experiments.

Index Terms—Magnetic equivalent circuit (MEC), mesh analysis, PM synchronous motor, reluctance, saturation.

I. INTRODUCTION

Design phase of electric machine undergoes repetitive analysis and involves change of machine parameters. The design has to fulfill multiple objective (like less mass to efficiency, low loss) by acknowledging every aspect like thermal, mechanical, control, besides its electromagnetic quantities, and hence algorithm takes longer execution time in converging to a design. So during design phase, an efficient model including all machine effects to analyse the motor behaviour is crucial. MEC is a useful approach to get solution in short time with an acceptable accuracy [1]. Literature support the use of MEC in machine design in [2], during its initial phase, and some utilised for analysing motors as in [3]–[6]. The PMSM was analysed in [3]–[5] accounting saturation to evaluate torque at steady state, and in [6] instantaneous torque was found.

When the machine has geometric complexities and exhibit non linearity, then MEC consumes time during computation. Methods proposed in literature to have fast solution combines MEC with finite element method in [7], MEC and Fourier modeling in [8], and so on. For a surface mounted magnet SM with heavily parameterised gear geometry, MEC has been developed in [9] with linear analysis, and [10] presents a non-linear MEC for a wound field SM for vehicle application. Each approach captures different performance metrics of the machine, and are useful in different operating regions.

Analysis of electric machines by modeling technique is a challenging task when saturation in iron core is to be considered. Due to the presence of saturation, the MEC is non-linear and this can either be analysed by node analysis

or mesh analysis iteratively. With mesh analysis, the solution converges within few iterations rather than a node analysis as concluded in [1]. So, it is decided to make use of mesh analysis in this work. While rotor motion is incorporated, it changes the mesh configuration at subsequent rotor positions. So there is a need to have a simple network topology.

In this paper, performance parameters of PMSM are evaluated by developing a MEC. It includes magnetic saturation, models rotor motion using mesh approach. When rotor motion is modeled, mesh structure is retained the same at various positions, but considers variation in magnetic coupling occurring during movement. Motor quantities solved by such model then provide significant information for designers to modify design at its early stage.

II. BUILDING MAGNETIC EQUIVALENT CIRCUIT

The cross section of PMSM considered to develop MEC is shown in Fig. 1. Stator has 12 slots and three phase winding with each phase comprising two coils connected in series. The coils of stator phase winding span over a slot pitch, and such configuration is commonly known as non-overlapping concentrated winding. There are 10 magnet poles placed on the surface of rotor. The complete cross section of motor is

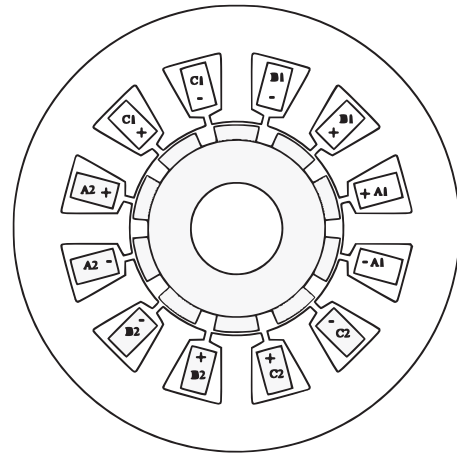


Fig. 1: Cross section of a 12 slot, 10 pole surface-mounted PMSM

considered for building MEC of which a portion of the model that includes one north and south pole is shown in Fig. 2. The regions depicted in MEC are stator, air gap and rotor core. In MEC, regions where flux is distributed is mapped as flux tubes, and these are depicted through reluctance elements. Sources in MEC are stator coils and rotor magnets, represented by MMF sources.

MMF sources of stator coils are placed in the yoke, and their value is determined from the current distribution of these coils. The magnet MMF sources are modeled as, $F_m = H_c l_m$ using the magnet's coercive force H_c and magnet length l_m . In the reluctance elements, three types are used for different sections like iron core, magnet and air gap. They are flux dependent reluctance in core, position dependent reluctance for air gap and magnets, and constant reluctance in slot opening space.

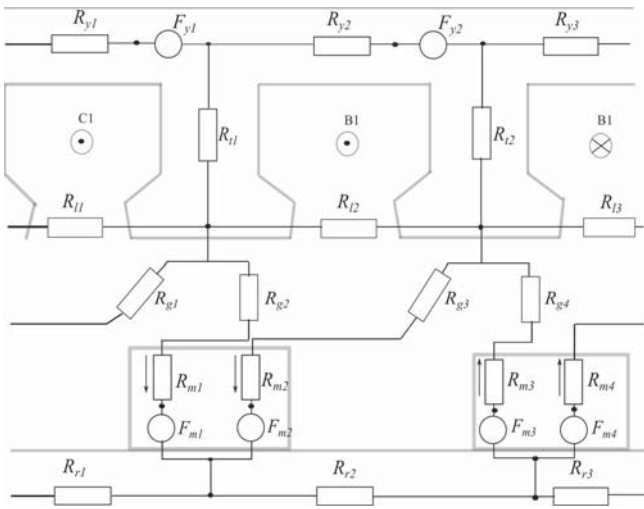


Fig. 2: Magnetic equivalent circuit

A. Representation of iron core

Core reluctance depends on material magnetisation curve whose properties are non-linear, and hence saturation effect in the core has to be accounted. To imbibe this feature in modeling, iron core reluctances are modeled as a function of flux, and is given as,

$$R_{s,j}(\phi) = \frac{l}{a \cdot \mu_0 \cdot \mu_r(\frac{\phi}{a})}, \quad j = 1, 2, 3, \dots, 12 \quad (1)$$

where l is mean length of the flux path, a cross sectional area, μ_0 permeability of free space, μ_r is flux dependent relative permeability. Here j represents the number of segments used in MEC. The shape of flux tubes considered to model stator core regions are approximated as rectangular element. For yoke and tooth regions, the modeling specified above is used. In Fig. 2, stator core reluctance R_s is constituted by R_y and R_t .

Rotor core reluctances' are modeled as illustrated in [11], and the relation used is,

$$R_{r,j} = \frac{1}{\mu_0 \mu_r(B)L} \left(\frac{l_2 - l_1}{h} \right) \frac{1}{\ln(l_2/l_1)}, \quad j = 1, 2, 3, \dots, 10 \quad (2)$$

where l is length of flux lines along the circumferential direction, h is mean height measured radially outward, and L is effective stack length. Rotor core flux tubes in this paper, are modeled as trapezoidal segments.

B. Representation of magnet

Magnets are represented as MMF sources (F_m) and reluctances (R_m), as mentioned. During rotation, flux from the magnet links the stator tooth, and the percentage of this flux varies with rotor position. This contribution of magnet flux for overall torque can be attributed either through F_m or R_m . In this paper, it is done by choosing suitable value of magnet reluctance. The magnet reluctance is calculated using the overlap area of magnet with the stator tooth it faces. To explain this, two magnet positions within a slot pitch (i.e., distance between centers of adjacent slots) are considered. When a complete magnet pole is enclosed in a slot pitch its reluctance is minimum to allow entire flux from magnet MMF. If a segment of the magnet is within slot pitch, its magnet reluctance is appropriately represented to indicate fraction of total flux linking the corresponding stator tooth. This explains magnet reluctance are a function of rotor position and is evaluated as,

$$R_{m,j}(\theta) = \frac{l}{a(\theta) \cdot \mu_0 \cdot \mu_r} \quad j = 1, 2, 3, \dots, 24 \quad (3)$$

where l is length of the magnet along which flux links, a its cross sectional area depends on rotor position, μ_0 permeability of free space, μ_r is relative permeability of magnet.

C. Representation of air gap

Air gap reluctance elements establishes connection between the stator and rotor for a given position. Whenever the rotor moves past a stator tooth, the connection in the air gap changes and hence this alters the MEC topology. For this, a common method in [11] was to connect each stator tooth with all rotor magnets by the air gap link. In this work, two reluctance branches from each stator tooth are used to connect with chosen magnet pole of the rotor. This branch element connection between the tooth and a pole is decided by the relative flux linkage existing between them and such neighbouring combinations. To do this, at the present rotor step, air gap reluctance between the facing stator tooth and magnet is compared with that of adjacent tooth-magnet, and whichever exhibits low reluctance will have the branch element coupling. While doing so, each of this branch ensures to connect with rotor pole of different magnetisation direction. A sample connection of air reluctance for a position is shown in Fig. 2.

Therefore air gap reluctance as a function of rotor position is calculated using expression in [11] as,

$$R_{g,n} = \begin{cases} R_{min} & 0 \leq \theta_m \leq \theta_{t1} \text{ and} \\ & 2\pi - \theta_{t1} \leq \theta_m \leq 2\pi \\ 2R_{min}/\left(1 + \cos\left(\pi \frac{\theta_m - \theta_{t1}}{\theta_{t2} - \theta_{t1}}\right)\right) & \theta_{t1} \leq \theta_m \leq \theta_{t2} \\ 2R_{min}/\left(1 - \cos\left(\pi \frac{\theta_m - 2\pi + \theta_{t2}}{\theta_m - \theta_{t1}}\right)\right) & 2\pi - \theta_{t2} \leq \theta_m \\ & \leq 2\pi - \theta_{t1} \\ R_{max} & \theta_{t2} \leq \theta_m \leq 2\pi - \theta_{t2} \end{cases} \quad (4)$$

where R_{min} is the minimum value corresponding to aligned position of tooth and rotor pole evaluated using $\frac{l}{\mu_0 \mu_r a}$, and R_{max} is chosen sufficiently high value when tooth and pole are completely out of alignment. θ_m is the angle (in mechanical measure) describing the reluctance variation in the interval $(0, \pi/6)$ mapped over a slot pitch. Further, θ_{t1} is the angle during which the tooth and pole remains in overlap, and from θ_{t1} to θ_{t2} reluctance varies from minimum to maximum value. In order to get smooth variation in reluctance, the effect of fringing is accounted.

D. Representation of slot leakage

Slot leakage constitutes to large part of leakage in motor, and has to be accounted in modeling. In some cases, slot leakage flux can influence saturation in adjacent teeth. For the slot shape shown in Fig. 3 having a single layer winding, leakage inductance given in [11] is used to find energy stored in the slot space. Then energy stored is expressed in terms of reluctance from which circumferential slot leakage reluctance is expressed as,

$$R_{l,j} = \frac{1}{2\mu_0 L \left(\frac{h_1}{b_1} + \frac{h_4}{3b_4} \right)}, \quad j = 1, 2, \dots, 12 \quad (5)$$

These reluctance elements (R_l) are placed in the slot space as shown in Fig. 2, and is modeled by linear reluctance. The variables in above equation can be referred from Fig. 3.

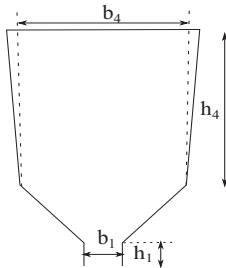


Fig. 3: Stator slot dimension

III. SOLUTION METHODOLOGY

A. Model equation

After formulating MEC, network equations for Fig. 2 will be of the form, $\mathbf{R}\Phi = \mathbf{F}$, where $\mathbf{R}$ is a symmetric reluctance matrix, Φ is the flux vector denotes loop fluxes, $\mathbf{F}$ is source MMF vector. These equations were formed by adopting mesh

approach, a feature significant for faster convergence as in [1]. Solution is obtained using Newton Raphson method, and the flow chart of overall procedure is given in Fig. 4.

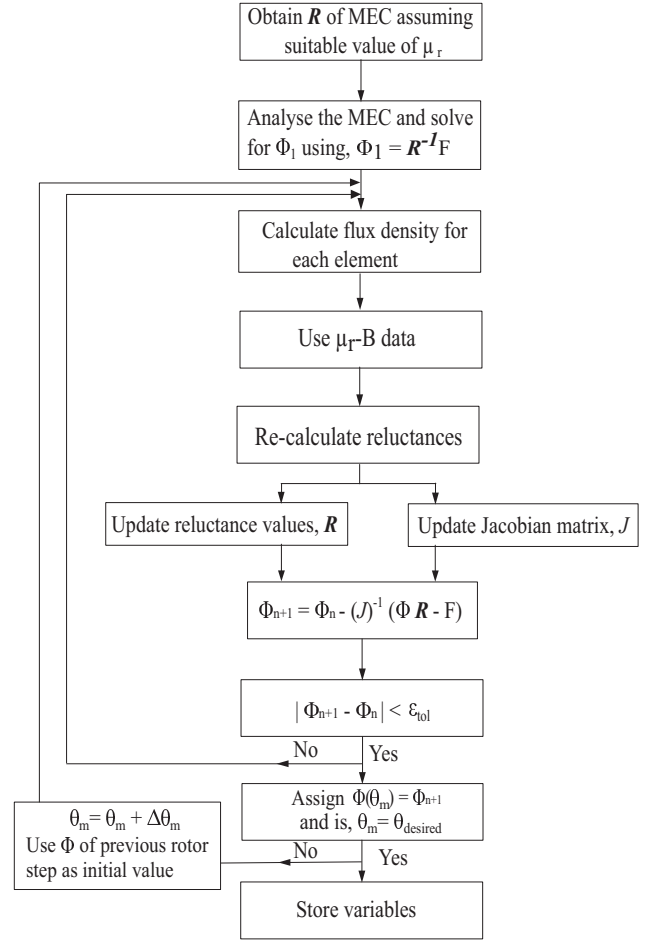


Fig. 4: Flowchart of solution procedure

Solution of MEC utilises input parameters like machine geometry, material properties and winding arrangement. To solve for flux, a maximum value of relative permeability is assumed to get reluctances assuming some initial rotor position θ_m . For each position, permeabilities of previous position are used to calculate reluctances. Calculation for next rotor step proceeds by obtaining flux density for each segment. Now μ_r and $\partial \mu_r / \partial B$ is determined from B- μ_r curve. Reluctance matrix $\mathbf{R}$ and Jacobian matrix are updated to get the flux value using,

$$\Phi_{n+1} = \Phi_n - \mathbf{J}_n^{-1} (\mathbf{R}_n \Phi_n - \mathbf{F}_n) \quad (6)$$

where $\mathbf{J}_n$, Jacobian matrix, derivative of reluctance matrix is calculated as described in [1], n indicates iteration number. This iterative procedure continues until the solution converges, such that $|\phi_{n+1} - \phi_n| < \epsilon$, where ϵ is the tolerance value. After convergence in one step angle, the rotor step angle $\Delta \theta_m$ is incremented, and same procedure is repeated until the required rotor angle. In every step angle, the value of ϕ from previous iteration is used to make convergence faster. So

the maximum number of iterations utilised here is less than three.

B. Validation of MEC

MEC quantities are validated using FEA, and the machine specifications used to develop MEC are listed below in Table. I.

TABLE I: PMSM Specification

Parameter	Value	Parameter	Value
<i>Performance</i>			
Rated power	1.5 kW	Rated torque	19 N-m
Rated speed	800 rpm	Voltage	48 V_{ll}
<i>General</i>			
Number of poles	10	Number of stator slots	12
Number of phases	3	Number of turns per phase	72
Stator outer diameter	146 mm	Stator inner diameter	70 mm
Rotor outer diameter	68 mm	Shaft diameter	30 mm
Air gap length	1 mm	Stack length	55 mm
Magnet thickness	5 mm	Pole-arc to pole-pitch ratio	0.7
Magnet remanence	1.1 T	Magnet permeability	1.04

Performance parameters evaluated from MEC are namely flux distribution, flux linkage with the stator winding, induced EMF, inductance and cogging torque. For these results, only magnets are considered to be the excitation source and the phase windings are unenergised. But phase inductance is obtained considering the stator coils also to be excited. To simulate the performance, an initial reference position is considered wherein, a rotor magnet axis coincides with phase A axis. Then it is rotated from its reference position to 72° mechanical angle at 1000 rpm and takes 144 discrete steps to produce an electrical cycle of waveform. The results presented in FEA uses ANSYS software in which time stepping simulation is performed. But for cogging torque magnetostatic solver is used in order to get precise meshing. Experimental results are presented for few quantities like phase flux linkages, induced EMF and inductance.

1) *Air gap flux density distribution*: Flux density distribution in the air gap obtained from MEC and FEA are presented in Fig. 5. This distribution obtained due to magnet flux agrees with that of FEA. The peak of flux density distribution evaluated from MEC either remains constant or inclined, and to get actual variation slot effects of the stator has to be accounted. In the subsequent work, slot modeling shall be considered.

2) *Flux linkage and Induced EMF*: Flux linkage calculation in MEC uses the details of winding layout, tooth flux and is shown in Fig. 6. The plot of flux linkage shown from experiment is derived from measured value of induced EMF. Also measured and calculated induced EMF are shown in Fig. 7. There is a good agreement among MEC, FEA and measurement, and error is within 5% for flux linkages and

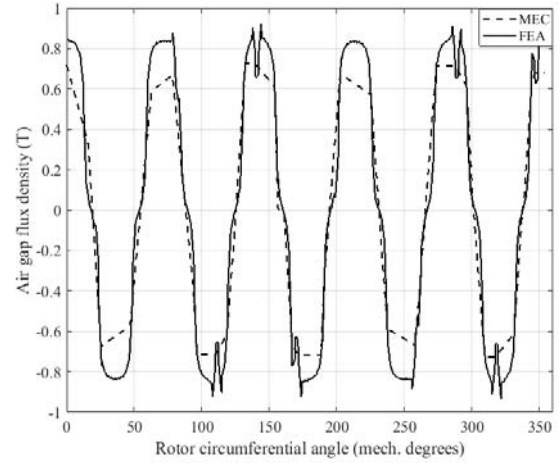


Fig. 5: Air gap flux density without stator excitation

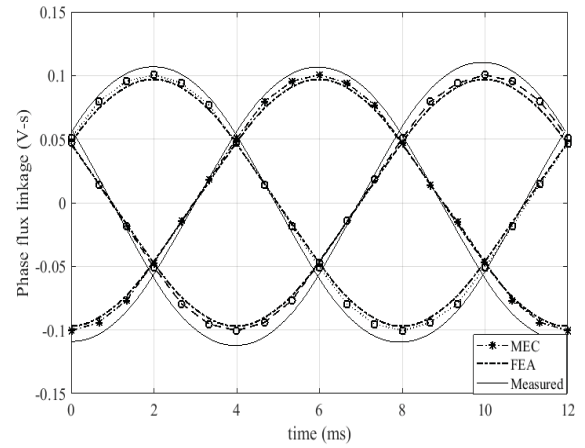


Fig. 6: Phase flux linkage

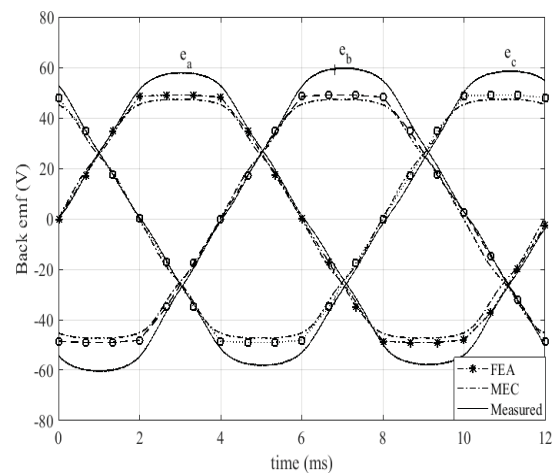


Fig. 7: Induced EMF

induced EMF.

3) *Phase Inductance*: The self inductance of a phase winding is given by the ratio of flux linkage with the winding to its current flow. Due to the symmetry between each phase, self inductance is calculated for only one phase. The self inductance of winding includes slot leakage effect. From MEC, self inductance of phase A winding is calculated as,

$$L_A = \frac{\lambda_{ct} - \lambda_{pm}}{i_A} \quad (7)$$

where λ_{ct} and λ_{pm} are the total flux linkage of the winding and flux linkage due to magnets respectively, i_A is current in phase A. The average of inductance L_A of phase A obtained from MEC, FEA and experimental measurement is given in Table. II.

TABLE II: Inductance calculation

Quantity	MEC	FEA	Experiment
L_A	5.94 mH (includes slot leakage)	6.33 mH (includes slot leakage)	6.77 mH (includes slot leakage and end winding effect)

As the PMSM has concentrated winding, the effect of end winding leakage is less, and hence the deviation observed in inductance measurement is negligible.

4) *Cogging torque*: Cogging torque caused by the interaction of permanent magnet with the stator slots is computed by applying change in co-energy principle. When co-energy is expressed as a function of flux and air gap reluctance, torque is obtained as,

$$T = \left(\frac{P}{2}\right) \sum_{j=1}^{n_g} \phi_g^2 \frac{\Delta R_g}{\Delta \theta_e} \quad (8)$$

where P is the pole number, n_g is number of air gap reluctances changing with position, ϕ_g is air gap flux of the corresponding reluctance element. Here, $\Delta \theta_e$ corresponds to incremental variation in rotor angle for which change in reluctance is computed. The cogging torque obtained from this

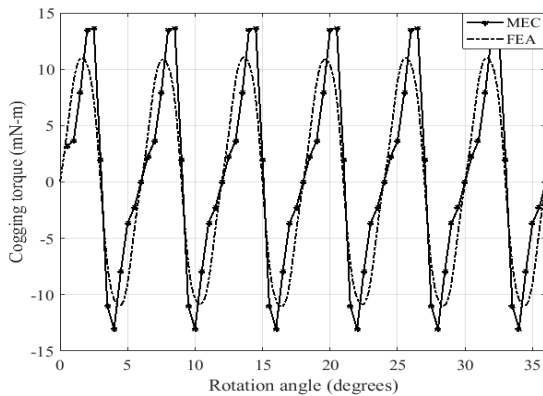


Fig. 8: Cogging torque

evaluation is compared with FEA and is shown in Fig. 8. The number of cogging cycles for the PMSM is 60 per revolution, and plot is shown for a rotation angle of 36° . A slightly higher estimate in cogging torque in MEC is observed because the magnet leakage flux was unaccounted during modeling.

IV. CONCLUSION

In this paper, MEC for fractional-slot PMSM is presented. The number of air gap branch elements used in the model minimises the dense network. The choice in air gap reluctances do not require reconstruction in circuit configuration at every step angle, even when, the relative position of rotor changes with respect to stator. This in turn facilitates solution by mesh approach with ease, and use of Newton Raphson method makes it computationally efficient. Comparison of evaluated quantities like flux density, flux linkage, induced EMF from MEC with FEA shows an error within 5%, and hence model can be useful in initial phase of design evaluations.

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